

# BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

**Name:** Parker Catena, Regina Velasco

**ID:** pdc76, rv297

## **Due Date**

Thursday, 10/16/25, 9:00pm

## **Overview**

### **Instructions**

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

### **Load Environment**

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/BEE 4750 /HW 3/hw3-parker`
Precompiling project...
 466.1 ms  WorkerUtilities
 472.1 ms  IteratorInterfaceExtensions
 471.7 ms  TensorCore
```

|           |                                        |
|-----------|----------------------------------------|
| 536.2 ms  | LaTeXStrings                           |
| 469.3 ms  | Contour                                |
| 478.7 ms  | StatsAPI                               |
| 356.6 ms  | Measures                               |
| 918.2 ms  | Format                                 |
| 479.9 ms  | Statistics                             |
| 503.9 ms  | OrderedCollections                     |
| 653.6 ms  | Grisu                                  |
| 302.9 ms  | DataValueInterfaces                    |
| 786.8 ms  | FillArrays                             |
| 443.3 ms  | Requires                               |
| 360.2 ms  | StableRNGs                             |
| 443.3 ms  | Unzip                                  |
| 310.8 ms  | Reexport                               |
| 502.0 ms  | DocStringExtensions                    |
| 378.1 ms  | SimpleBufferStream                     |
| 396.2 ms  | PtrArrays                              |
| 612.3 ms  | InlineStrings                          |
| 612.1 ms  | URIs                                   |
| 1750.5 ms | MacroTools                             |
| 411.4 ms  | InvertedIndices                        |
| 1029.5 ms | IrrationalConstants                    |
| 499.4 ms  | NaNMath                                |
| 475.3 ms  | DelimitedFiles                         |
| 616.1 ms  | TranscodingStreams                     |
| 421.0 ms  | DataAPI                                |
| 355.2 ms  | BitFlags                               |
| 376.4 ms  | Scratch                                |
| 720.3 ms  | ConcurrentUtilities                    |
| 982.5 ms  | Crayons                                |
| 598.2 ms  | LoggingExtras                          |
| 904.4 ms  | MbedTLS                                |
| 1071.9 ms | SentinelArrays                         |
| 754.2 ms  | Preferences                            |
| 624.8 ms  | Compat                                 |
| 579.7 ms  | ExceptionUnwrapping                    |
| 1521.8 ms | PDMats                                 |
| 613.5 ms  | TableTraits                            |
| 704.5 ms  | Statistics → SparseArraysExt           |
| 401.9 ms  | Showoff                                |
| 1210.5 ms | UnicodeFun                             |
| 671.3 ms  | FillArrays → FillArraysStatisticsExt   |
| 637.4 ms  | FillArrays → FillArraysSparseArraysExt |

|           |                                  |
|-----------|----------------------------------|
| 567.4 ms  | AliasTables                      |
| 791.2 ms  | CodecZlib                        |
| 1826.4 ms | FixedPointNumbers                |
| 1596.3 ms | DataStructures                   |
| 868.4 ms  | Missings                         |
| 935.2 ms  | PooledArrays                     |
| 1450.9 ms | LogExpFunctions                  |
| 958.0 ms  | RelocatableFolders               |
| 750.2 ms  | PrecompileTools                  |
| 767.0 ms  | JLLWrappers                      |
| 886.9 ms  | Compat → CompatLinearAlgebraExt  |
| 1105.7 ms | FillArrays → FillArraysPDMatsExt |
| 1338.0 ms | SortingAlgorithms                |
| 1431.5 ms | QuadGK                           |
| 1776.2 ms | ColorTypes                       |
| 2331.3 ms | Tables                           |
| 742.6 ms  | OpenSSL_jll                      |
| 529.5 ms  | Graphite2_jll                    |
| 532.9 ms  | Libmount_jll                     |
| 2046.7 ms | RecipesBase                      |
| 1927.2 ms | StringManipulation               |
| 435.2 ms  | EpollShim_jll                    |
| 412.7 ms  | LLVMOpenMP_jll                   |
| 432.3 ms  | Bzip2_jll                        |
| 391.7 ms  | Xorg_libICE_jll                  |
| 430.6 ms  | Rmath_jll                        |
| 397.4 ms  | Xorg_libXau_jll                  |
| 416.3 ms  | libfdk_aac_jll                   |
| 421.7 ms  | libpng_jll                       |
| 438.8 ms  | LAME_jll                         |
| 397.3 ms  | fzf_jll                          |
| 406.7 ms  | LERC_jll                         |
| 507.5 ms  | JpegTurbo_jll                    |
| 472.8 ms  | Ogg_jll                          |
| 508.2 ms  | XZ_jll                           |
| 365.1 ms  | mtdev_jll                        |
| 382.9 ms  | Xorg_libXdmcp_jll                |
| 419.3 ms  | x265_jll                         |
| 455.0 ms  | x264_jll                         |
| 441.8 ms  | Zstd_jll                         |
| 455.9 ms  | libaom_jll                       |
| 434.1 ms  | Expat_jll                        |
| 424.8 ms  | LZO_jll                          |

|           |                                        |
|-----------|----------------------------------------|
| 440.0 ms  | Opus_jll                               |
| 403.7 ms  | Xorg_xtrans_jll                        |
| 386.6 ms  | libevdev_jll                           |
| 421.6 ms  | Libiconv_jll                           |
| 414.1 ms  | Libffi_jll                             |
| 412.5 ms  | eudev_jll                              |
| 390.3 ms  | Libuuid_jll                            |
| 658.7 ms  | OpenSpecFun_jll                        |
| 416.4 ms  | FriBidi_jll                            |
| 588.2 ms  | FilePathsBase                          |
| 548.8 ms  | ColorTypes → StyledStringsExt          |
| 1305.8 ms | Pixman_jll                             |
| 2697.8 ms | ColorVectorSpace                       |
| 445.8 ms  | Xorg_libSM_jll                         |
| 1275.4 ms | FreeType2_jll                          |
| 3301.6 ms | OpenSSL                                |
| 1080.2 ms | Rmath                                  |
| 1350.9 ms | JLFzf                                  |
| 4577.8 ms | Colors                                 |
| 4931.8 ms | StatsBase                              |
| 1324.6 ms | Ghostscript_jll                        |
| 720.4 ms  | libvorbis_jll                          |
| 358.8 ms  | Xorg_libxcb_jll                        |
| 8681.3 ms | Parsers                                |
| 380.1 ms  | DBus_jll                               |
| 373.2 ms  | Wayland_jll                            |
| 350.9 ms  | libinput_jll                           |
| 876.7 ms  | GettextRuntime_jll                     |
| 924.3 ms  | FilePathsBase → FilePathsBaseMmapExt   |
| 1281.5 ms | FilePathsBase → FilePathsBaseTestExt   |
| 1828.1 ms | Libtiff_jll                            |
| 311.3 ms  | Xorg_xcb_util_jll                      |
| 1902.6 ms | SpecialFunctions                       |
| 307.1 ms  | Xorg_libX11_jll                        |
| 2611.8 ms | Fontconfig_jll                         |
| 1728.4 ms | JSON                                   |
| 316.3 ms  | Xorg_xcb_util_image_jll                |
| 1792.6 ms | InlineStrings → ParsersExt             |
| 363.8 ms  | Xorg_xcb_util_keysyms_jll              |
| 370.3 ms  | Xorg_xcb_util_renderutil_jll           |
| 441.5 ms  | Xorg_xcb_util_wm_jll                   |
| 2041.7 ms | Glib_jll                               |
| 509.0 ms  | ColorVectorSpace → SpecialFunctionsExt |

|            |                                      |
|------------|--------------------------------------|
| 411.5 ms   | Xorg_libXrender_jll                  |
| 1155.3 ms  | HypergeometricFunctions              |
| 4498.8 ms  | ColorSchemes                         |
| 389.4 ms   | Xorg_libXext_jll                     |
| 474.1 ms   | Xorg_libXfixes_jll                   |
| 422.3 ms   | Xorg_libxkbfile_jll                  |
| 501.1 ms   | Xorg_xcb_util_cursor_jll             |
| 819.8 ms   | WeakRefStrings                       |
| 4950.5 ms  | Latexify                             |
| 453.8 ms   | Xorg_libXinerama_jll                 |
| 400.1 ms   | Xorg_libXrandr_jll                   |
| 376.6 ms   | Libglvnd_jll                         |
| 360.8 ms   | Xorg_libXi_jll                       |
| 368.8 ms   | Xorg_libXcursor_jll                  |
| 407.4 ms   | Xorg_xkbcomp_jll                     |
| 2207.4 ms  | StatsFuns                            |
| 1296.5 ms  | Latexify → SparseArraysExt           |
| 1099.6 ms  | Xorg_xkeyboard_config_jll            |
| 2962.5 ms  | Cairo_jll                            |
| 506.9 ms   | xkbcommon_jll                        |
| 565.2 ms   | Vulkan_Loader_jll                    |
| 1517.2 ms  | HarfBuzz_jll                         |
| 1971.9 ms  | libass_jll                           |
| 846.6 ms   | Pango_jll                            |
| 5234.3 ms  | Qt6Base_jll                          |
| 9496.1 ms  | PlotUtils                            |
| 465.5 ms   | libdecor_jll                         |
| 1608.7 ms  | Qt6ShaderTools_jll                   |
| 3645.9 ms  | FFMPEG_jll                           |
| 9047.8 ms  | Distributions                        |
| 566.3 ms   | GLFW_jll                             |
| 3991.1 ms  | PlotThemes                           |
| 1767.3 ms  | FFMPEG                               |
| 2572.4 ms  | Qt6Declarative_jll                   |
| 20109.4 ms | HTTP                                 |
| 588.2 ms   | Qt6Wayland_jll                       |
| 1508.2 ms  | Distributions → DistributionsTestExt |
| 1629.5 ms  | GR_jll                               |
| 4958.7 ms  | RecipesPipeline                      |
| 16889.5 ms | CSV                                  |
| 28853.5 ms | PrettyTables                         |
| 2785.7 ms  | GR                                   |
| 29396.6 ms | DataFrames                           |

```
1600.2 ms    Latexify → DataFramesExt
40598.2 ms   Plots
177 dependencies successfully precompiled in 81 seconds. 37 already precompiled.
```

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

## Problems (Total: 30 Points)

### Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day<sup>-1</sup>, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day<sup>-1</sup>, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance  $d$  km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

| Parameter        | River Inflow              | Waste Stream 1           | Waste Stream 2           |
|------------------|---------------------------|--------------------------|--------------------------|
| Inflow           | 100,000 m <sup>3</sup> /d | 10,000 m <sup>3</sup> /d | 15,000 m <sup>3</sup> /d |
| DO Concentration | 7.5 mg/L                  | 5 mg/L                   | 5 mg/L                   |
| CBOD             | 5 mg/L                    | 50 mg/L                  | 45 mg/L                  |
| NBOD             | 5 mg/L                    | 35 mg/L                  | 35 mg/L                  |

Table 1: River inflow and waste stream characteristics for Problem 1.

$$K_a = 0.55 \text{ per day}$$

$$K_c = 0.35 \text{ per day}$$

$$K_n = 0.25 \text{ per day}$$

$$C_s = 10 \text{ mg/L}$$

### Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```
u = 6 #km/day
dist = 15 #km
r = 0.55 #1/day
kc = 0.35 #1/day
kn = 0.25 #1/day
Cs = 10 #mg/L

Q_r0 = 100000 #m3/day
DO_r0 = 7.5 #mg/L
C_r0 = 5 #mg/L
N_r0 = 5 #mg/L

Q_w1 = 10000 #m3/day
DO_w1 = 5 #mg/L
C_w1 = 50 #mg/L
N_w1 = 35 #mg/L

Q_w2 = 15000 #m3/day
DO_w2 = 5 #mg/L
C_w2 = 45 #mg/L
N_w2 = 35 #mg/L

Q_out1 = Q_r0 + Q_w1

function conc_initial(c0, q0, c1, q1, qout)
    ((c0*q0) + (c1*q1)) / qout
end

DO_out1 = conc_initial(DO_r0, Q_r0, DO_w1, Q_w1, Q_out1)
C_out1 = conc_initial(C_r0, Q_r0, C_w1, Q_w1, Q_out1)
N_out1 = conc_initial(N_r0, Q_r0, N_w1, Q_w1, Q_out1)

function conc(x, u, r, kc, kn, D00, C0, N0, Cs)
    a1 = exp(-r*x/u)
    a2 = (kc/(r-kc))*(exp(-kc*x/u) - a1)
    a3 = (kn/(r-kn))*(exp(-kn*x/u) - a1)
    Cs*(1 - a1) + D00*a1 - C0*a2 - N0*a3
```

```

end

# UNTIL X = 15 RIGHT BEFORE SECOND WASTE STREAM INPUT
x1 = range(0.0, stop=15.0, length=1501)      # <- use 0..15 here
C1 = conc.(x1, u, r, kc, kn, DO_out1, C_out1, N_out1, Cs)

# values at x=15
# (L0 and N0 were undefined; you meant the mixed CBOD/NBOD after the first discharge)
L15 = C_out1 * exp(-kc*15.0/u)                # scalar math is fine
N15 = N_out1 * exp(-kn*15.0/u)
C15 = C1[end]                                # indexing uses [end], not C1(end)

# SECOND WASTE STREAM INPUT AT X = 15
Q_out2 = Q_out1 + Q_w2
DO_out2 = conc_initial(C15, Q_out1, DO_w2, Q_w2, Q_out2)
C_out2 = conc_initial(L15, Q_out1, C_w2, Q_w2, Q_out2)
N_out2 = conc_initial(N15, Q_out1, N_w2, Q_w2, Q_out2)

x2 = range(15.0, stop=50.0, length=3501)
C2 = conc.(x2 .- 15.0, u, r, kc, kn, DO_out2, C_out2, N_out2, Cs)

x_total = [x1; x2[2:end]]
C_total = [C1; C2[2:end]]
imin = argmin(C_total)
minDO = C_total[imin]
x_at_min = x_total[imin]
println("Minimum DO = $(round(minDO, digits=3)) mg/L at x = $(round(x_at_min, digits=3)) km")

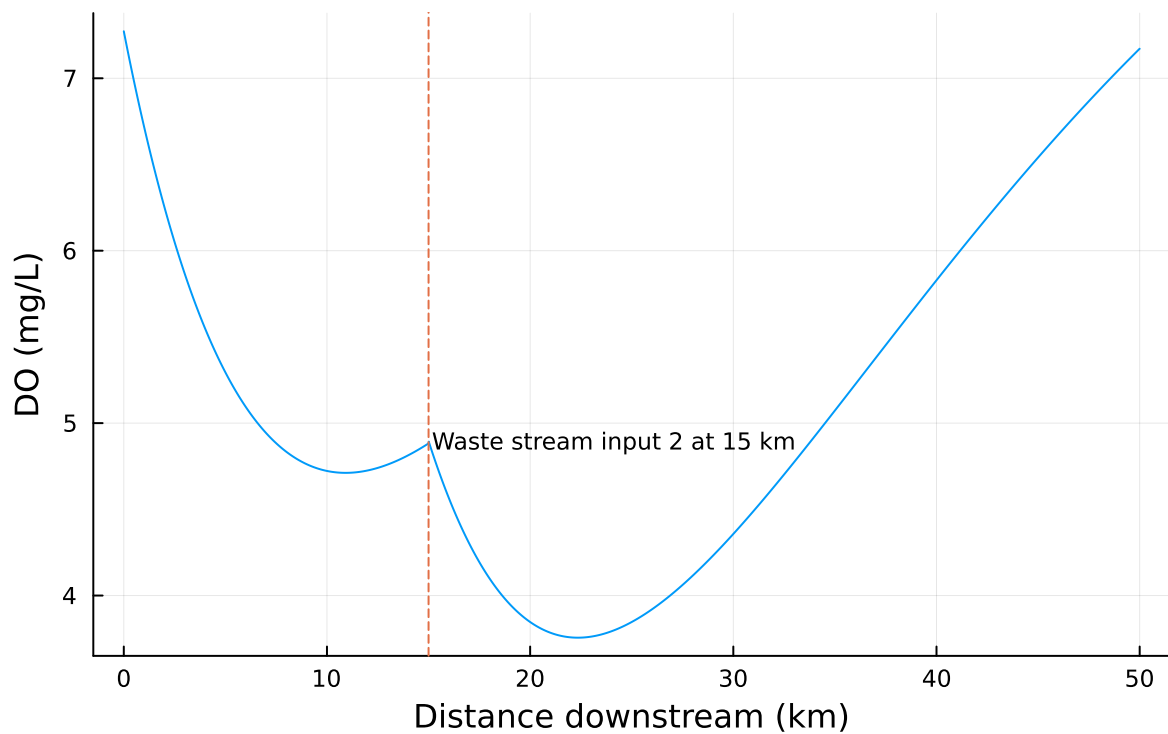
# plot
plot(x_total, C_total, xlabel = "Distance downstream (km)", ylabel = "DO (mg/L)",
      title = "DO Concentration Downstream with Two Waste Inputs", legend = false)
vline!([15.0], linestyle = :dash)
annotate!(15.2, DO_out2, text("Waste stream input 2 at 15 km", :left, 8))

```

Minimum DO = 3.756 mg/L at x = 22.35 km



## DO Concentration Downstream with Two Waste Inputs



### Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

```
dx = 0.5 #km
L = 50 #km

function do_numerical(L, dx, C0, B0, N0, ka, kn, kc, Cs, U) #Numerical implementation function
    n = Int(L / dx) # number of length steps
    C = zeros(n + 1)
    B = zeros(n + 1)
    N = zeros(n + 1)
    C[1] = C0
    B[1] = B0
    N[1] = N0
```

```

    for i = 1:n
        B[i+1] = B[i] * exp(-kc * dx / U)
        N[i+1] = N[i] * exp(-kn * dx / U)
        C[i+1] = C[i] + (dx / U) * (ka * (Cs - C[i]) - kc * B[i] - kn * N[i])
    end
    return (C, B, N)
end

#Right after first waste input
C1, B1, N1 = do_numerical(15.0, dx, DO_out1, C_out1, N_out1, r, kn, kc, Cs, u)

Q_out2 = Q_out1 + Q_w2
DO_15 = C1[end]
L_15 = B1[end]
N_15 = N1[end]

DO_mix2 = conc_initial(DO_15, Q_out1, DO_w2, Q_w2, Q_out2)
L_mix2 = conc_initial(L_15, Q_out1, C_w2, Q_w2, Q_out2)
N_mix2 = conc_initial(N_15, Q_out1, N_w2, Q_w2, Q_out2)

#Right after second waste input
C2, B2, N2 = do_numerical(35.0, dx, DO_mix2, L_mix2, N_mix2, r, kn, kc, Cs, u)

x = [0.0:dx:15.0; 15.0+dx:dx:50.0]
DO_tot = [C1; C2[2:end]]

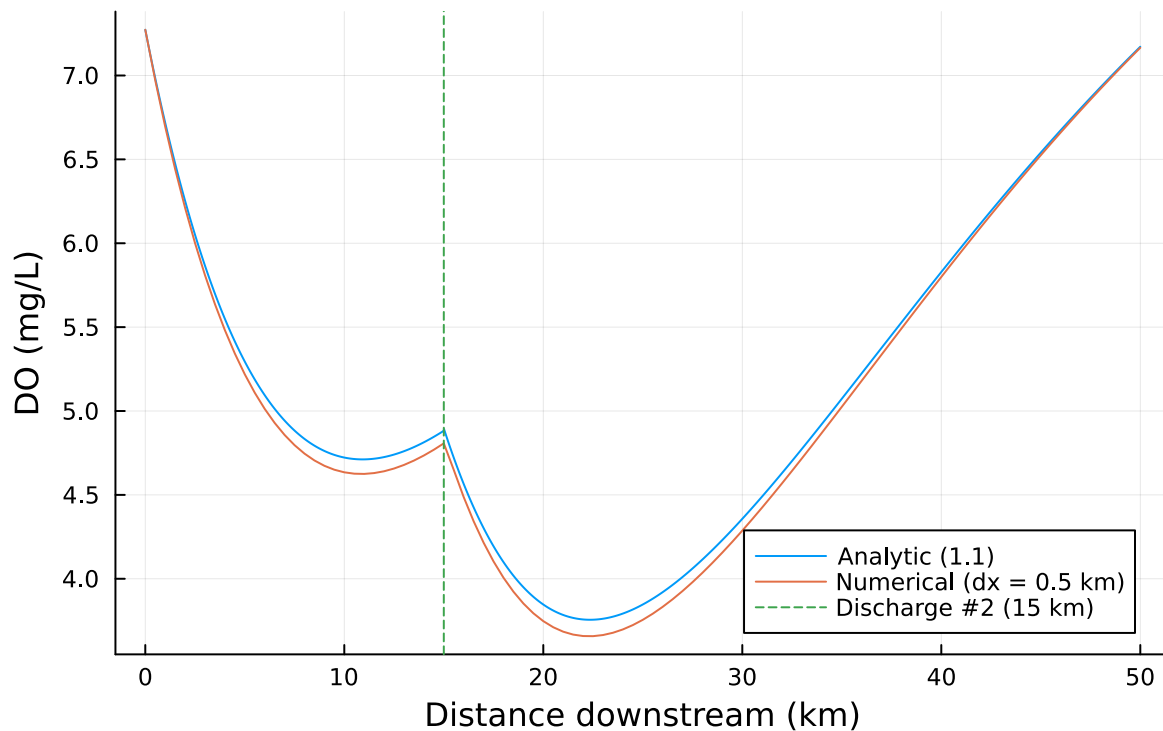
imin_num = argmin(DO_tot)
println("Numerical min DO = $(round(DO_tot[imin_num], digits=3)) mg/L at x = $(round(x[imin_num], digits=3)) km")

# Plot on same axes as 1.1 (if you computed x_all, C_all); otherwise just plot numerical.
if @isdefined(x_total) && @isdefined(C_total)
    plot(x_total, C_total, label="Analytic (1.1)", xlabel="Distance downstream (km)", ylabel="DO Concentration (mg/L)",
        title="DO Concentration: Analytic vs Numerical (dx = 0.5 km)")
    plot!(x, DO_tot, label="Numerical (dx = 0.5 km)")
    vline!([15.0], linestyle=:dash, label="Discharge #2 (15 km)")
else
    plot(x, DO_tot, label="Numerical (dx = 0.5 km)", xlabel="Distance downstream (km)", ylabel="DO Concentration (mg/L)",
        title="DO Concentration (Numerical, lecture scheme)")
    vline!([15.0], linestyle=:dash, label="Discharge #2 (15 km)")
end
end

```

Numerical min DO = 3.658 mg/L at x = 22.5 km

### DO Concentration: Analytic vs Numerical (dx = 0.5 km)



#### Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

#### Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

### **Problem 1.5**

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

### **References**

List any external references consulted, including classmates.